

$$\textcircled{1} \quad \alpha_{Th} = 10 \times 10^{-6} \quad \dot{f} = 8 \times 10^{13} \text{ fiss/cm}^3 \text{ s} \quad T_F = 1500 \text{ K}$$

$$T_{ref} = 300 \text{ K} \quad \Delta \rho_0 = 0.015 \quad B_D = 5 \text{ MWp/K}_2\text{V}$$

$$t = 300 \text{ days}$$

$$\rho(U_{0D}) = 10.977 \text{ g/cc}$$

$$\text{assum } Nu = 2.44 \times 10^{-22}$$

Total change in fuel volume

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$$\epsilon_{Tot} = \epsilon_{TE} + \epsilon_{SFP} + \epsilon_{GFP} + \epsilon_D$$

$$B = \frac{\dot{f} V}{Nu}$$

$$B = \frac{8 \times 10^{13} \times 300 \text{ days} \times 24 \text{ hr} \times 60 \text{ min} \times 60 \text{ sec}}{10.977 \text{ g/cc} \times 1 \text{ cm}^3 \times 2.44 \times 10^{-22}}$$

$$B = \frac{8 \times 10^{13} \times 2.592 \times 10^6}{10.977 \times 2.44 \times 10^{-22}} = 1.085$$

$$\epsilon_{SFP} = 5.577 \times 10^{-2} B$$

$$\epsilon_{GFP} = 1.96 \times 10^{-28} B (2800 - T)^{11.73} \times e^{-0.0162(2800 - T)} \times e^{-17.88 T}$$

$$\epsilon_D = \Delta \rho_0 \left(e^{\frac{B \ln \rho_0}{C_D B_D}} - 1 \right)$$

$$B_D = \frac{5}{970} = .005$$

$$B > B_D$$

$$\epsilon_{SFP} = 5.577 \times 10^{-2} \times 10.977 \times .085 = .052$$

$$\epsilon_{GFP} = 1.96 \times 10^{-28} \times 10.977 \times .085 (2800 - 1500)^{11.73} \times e^{-0.0162(2800 - 1500)} \times e^{-17.88 \times 10.977 \times .085}$$

$$.215 \times 2.857 \times 7.14 \times 10^{-10} \times 6.126 \times 10^{-8}$$

$$\epsilon_{GFP} = 2.68 \times 10^{-17}$$

$$\epsilon_D = .015 \left(\frac{.005 \cdot \ln \rho_0}{.005} - 1 \right)$$

$$\epsilon_D = -.084$$

?

$$B > B_D \rightarrow \epsilon_D = -0.015$$

$$\epsilon_{TE} = ?$$

②

Total creep

$$\sigma = 60 \text{ MPa}$$

$$T = 600 \text{ K}$$

$$LHA = 250 \text{ W/cm}$$

$$t = 200 \text{ days}$$

$$\Lambda_0 = 3.14 \times 10^{24}$$

$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T \text{ Pa}$$

$$n = 5$$

$$Q = 2.7 \times 10^5$$

$$C_0 = 1.654 \times 10^{-24}$$

$$C_1 = .85$$

$$C_2 = 1$$

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$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 \cdot 600$$

$$G = 2.92 \times 10^{10} \text{ (Pa)}$$

$$G = 2.92 \times 10^{10}$$

$$\dot{\epsilon}_{th} = 3.14 \times 10^{24} \left(\frac{60 \text{ MPa}}{2.92 \times 10^{10}} \right)^5$$

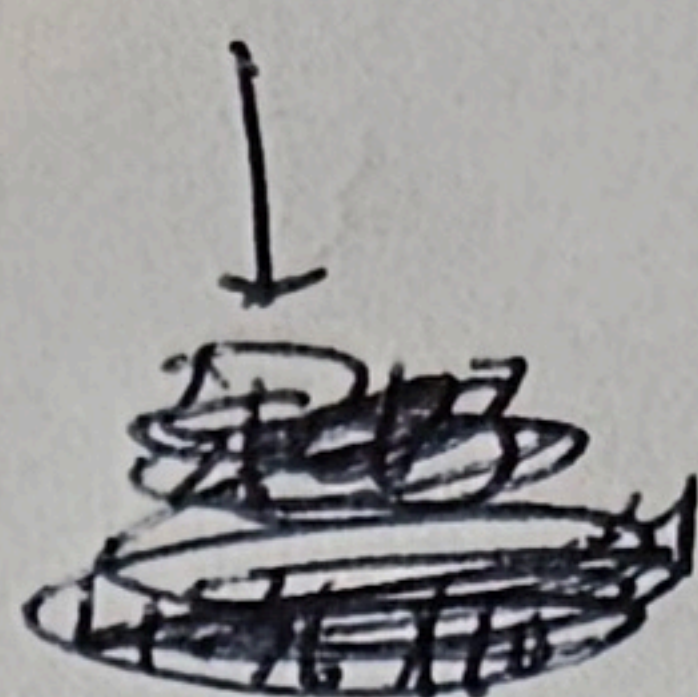
$$e^{\left(\frac{-2.7 \times 10^5}{8.314 \cdot 600} \right)}$$

$$\downarrow$$

$$3.11 \times 10^{-24}$$

What happened with units here?

$$\dot{\epsilon}_{th} =$$



$$\dot{\epsilon}_{th} = \frac{3.11 \times 10^{-24}}{3.16 \times 10^{-9}}$$

$$\dot{\epsilon}_{ir} = 1.654 \times 10^{-24} \cdot (7.5 \times 10^{13})^{.85} \cdot 60^1$$

$$\downarrow$$

$$6.22 \times 10^{-11}$$

$$\dot{\epsilon}_{ir} = 6.17 \times 10^{-8}$$

$$\frac{200 \text{ days} / 24 \text{ hr} / 60 \text{ min} / 60 \text{ sec}}{1} = 1728 \times 10^4$$

- I got a bit lost here...

$$\dot{\epsilon}_{th} = \frac{3.11 \times 10^{-24}}{3.16 \times 10^{-9}} \cdot 618$$

$$\dot{\epsilon}_{ir} = 1.066$$

$$\dot{\epsilon}_{T-H1} = 7.246$$

- this is a LOT!

③

250 days 14 00 K

$$\dot{F} = 2 \times 10^{13}$$

$$a_1 = 5 \text{ microns}$$

$$a_2 = 40 \text{ microns}$$

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$$\tau_1 = \frac{D_1}{a_1^2} = 4.619 \times 10^{-16}$$

$$\tau_1 = \frac{9.99704 \times 10^{-9}}{(5 \times 10^{-5})^2}$$

$$\tau_1 = 3.99 \times 10^{-4}$$

$$\tau_2 = \frac{9.99704 \times 10^{-9}}{(40 \times 10^{-5})^2}$$

$$\tau_2 = 6.24 \times 10^{-6}$$

$$\tau_1, \tau_2 \propto \tau^{-2}$$

$$f_{1,2} = 4 \sqrt{\frac{9.99704 \times 10^{-9}}{\tau^2}} - \frac{3}{2} \frac{9.99704 \times 10^{-9}}{\tau^2}$$

$$f_1 = 4 \sqrt{\frac{9.99704 \times 10^{-9}}{(3.99 \times 10^{-4})^2}} - \frac{3}{2} \frac{9.99704 \times 10^{-9}}{(3.99 \times 10^{-4})^2}$$

$$f_1 = 4 \sqrt{\frac{3.998816 \times 10^{-4}}{1}} - \frac{3}{2} \cdot 3.998816 \times 10^{-4}$$

↓ ↓

.045 5.999224 × 10⁻⁴

$$f_1 = .0445 = \boxed{4.45\%}$$

$$f_2 = 4 \sqrt{\frac{9.99704 \times 10^{-9}}{.0016}} - \frac{3}{2} \frac{9.99704 \times 10^{-9}}{.0016}$$

↓ ↓

.0056 9.3728475 × 10⁻⁶

$$f_2 = .0056 = \boxed{.56\%}$$

$$t = 250 \cdot 24 \cdot 60 \cdot 60$$

$$t = 216 \times 10^5$$

$$D_1 = 9.99704 \times 10^{-9}$$

D is cm²/s

units gives

all good, except units!

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.08}{T_{RT}}\right)$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.08}{8.617 \times 10^{-5} \cdot 1400}\right)$$

$$D_1 = 9.38 \times 10^{-17}$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{8.617 \times 10^{-5} \cdot 1400}\right) \sqrt{240^3}$$

$$5.2 \times 10^{-5}$$

$$D_2 = 3.28 \times 10^{-16}$$

$$D_3 = 2 \times 10^{-30} \cdot 2 \times 10^{13}$$

$$D_3 = 4 \times 10^{-17}$$

$$D = 4.619 \times 10^{-16}$$

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Typical charge of U in UO_2 is 4+ ✓

It can be changed due to charged defects entering the lattice or oxygen leaving the lattice. It can change its valence to be any of 3+, 4+, 5+, 6+ ✓

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⑤ Oxygen concentration is relatively constant across the fuel pellet. ✓

except @ periphery

Mo oxidizes with oxygen to fix the oxygen content across the fuel pellet

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⑥ Thermal conductivity, fuel melt temperature, lattice parameter

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⑦ Soluble oxides, insoluble oxides, metal inclusions, volatiles, noble gases

soluble oxides → increase swelling, reduce thermal conductivity

insoluble oxides → reduce thermal conductivity

metal inclusions → increase thermal conductivity

volatile → reduce thermal conductivity, increase swelling, corrode cladding, reduce gap thermal conductivity, increase gap pressure

noble gases → reduce thermal conductivity, increase swelling, reduce gap thermal conductivity, increase gap pressure

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⑧ Stage 1: Gas bubbles formed from fission. Are in the grain, diffuse toward grain boundary ✓

Stage 2: Gases accumulate in GB and form GB bubbles

Stage 3: GB bubbles interconnect and migrate via GB's to pellet edge and escape

⑨ + Creep is the deformation of a solid with no change ~~in~~ volume when $\sigma < \sigma_y$. This is caused by the movement of dislocations.

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- Nabarro-Herring creep. Atoms diffuse, Grain boundaries grow along load direction. ~~Sometimes~~

⑩ The HBS is the outside radius of the fuel pellet that recrystallizes due to the increase in damage to the rim structure. As fission occurs and neutrons are moderated, epithermal neutrons are preferentially absorbed by ^{238}U in the rim leading to the production of ~~of~~ Pu-239 . This causes an overall increase in fission in the rim, causing an increase in the production of defects in the rim. The recrystallization that occurs because of this sweeps defects out of the grains as grain size is reduced, and leads to a highly porous structure that traps fission gasses. ~~the~~ Although defect concentrations are higher in the rim, the defect free grains cause a slight increase in overall thermal conductivity. The trapping of the ~~of~~ fission gasses also limits gas thermal conductivity reduction, limits the increase in gap pressure, and reduces volatile gas concentration that would corrode the cladding.

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⑪

Two mechanisms cause defect induced growth in zirconium.

1: Interstitial defects will form loops on the ~~prismatic~~ plane. This causes an increase in the ~~a~~ direction and a decrease in the ~~z~~ direction.

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2: Vacancies will form on the basal plane, ~~once again~~ causing an increase in the ~~a~~ direction and a decrease in the ~~z~~ direction.

(14)

Iodine combines with zinc on a crack face. ZnI_2 then migrates up the temperature gradient where the Zn combines with free oxygen to form ZnO . This can form on the walls of the crack forming a protective layer. The free iodine then travels back down the temperature gradient and attacks the crack face, recombining with zinc and the cycle happens again.

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